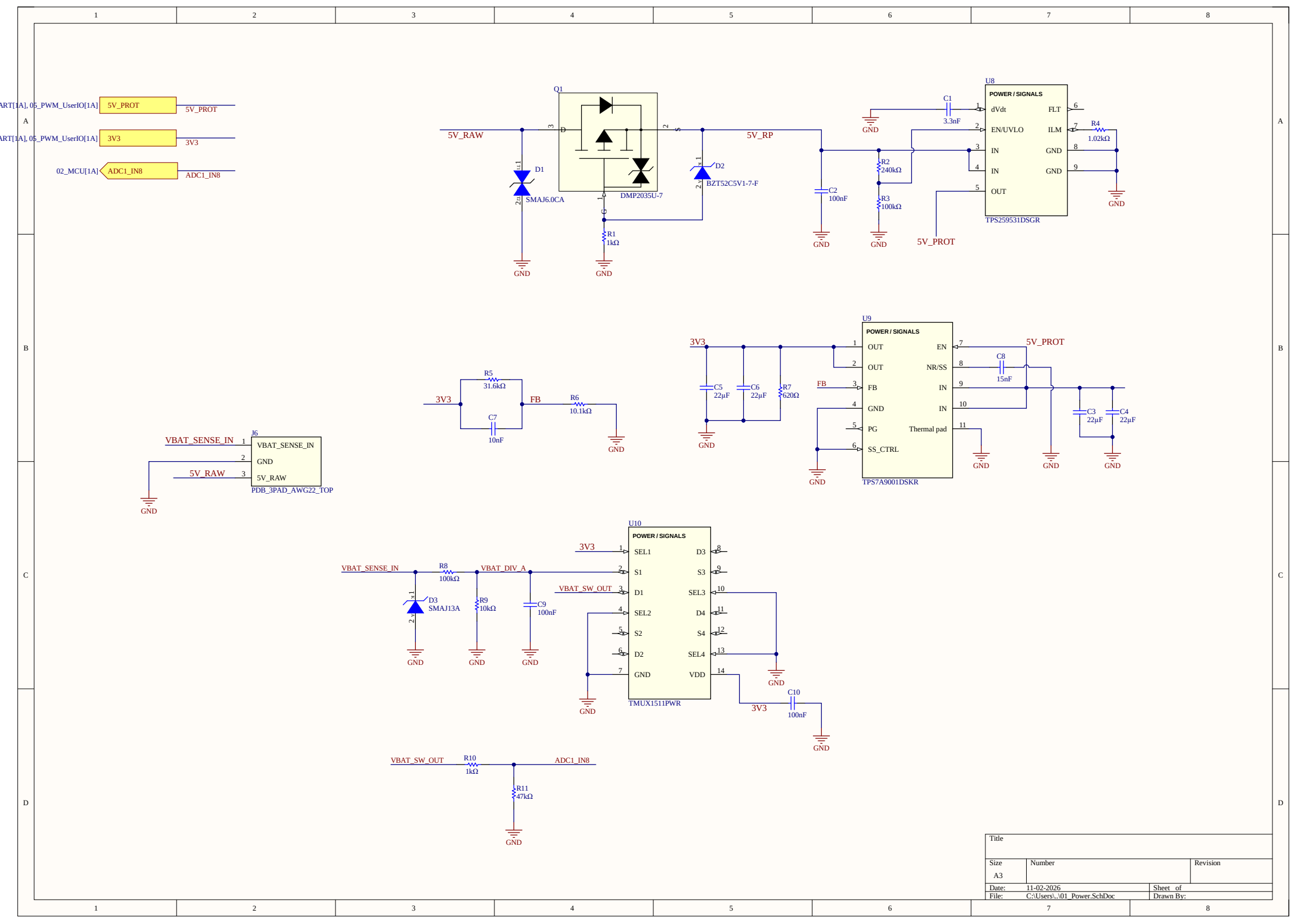
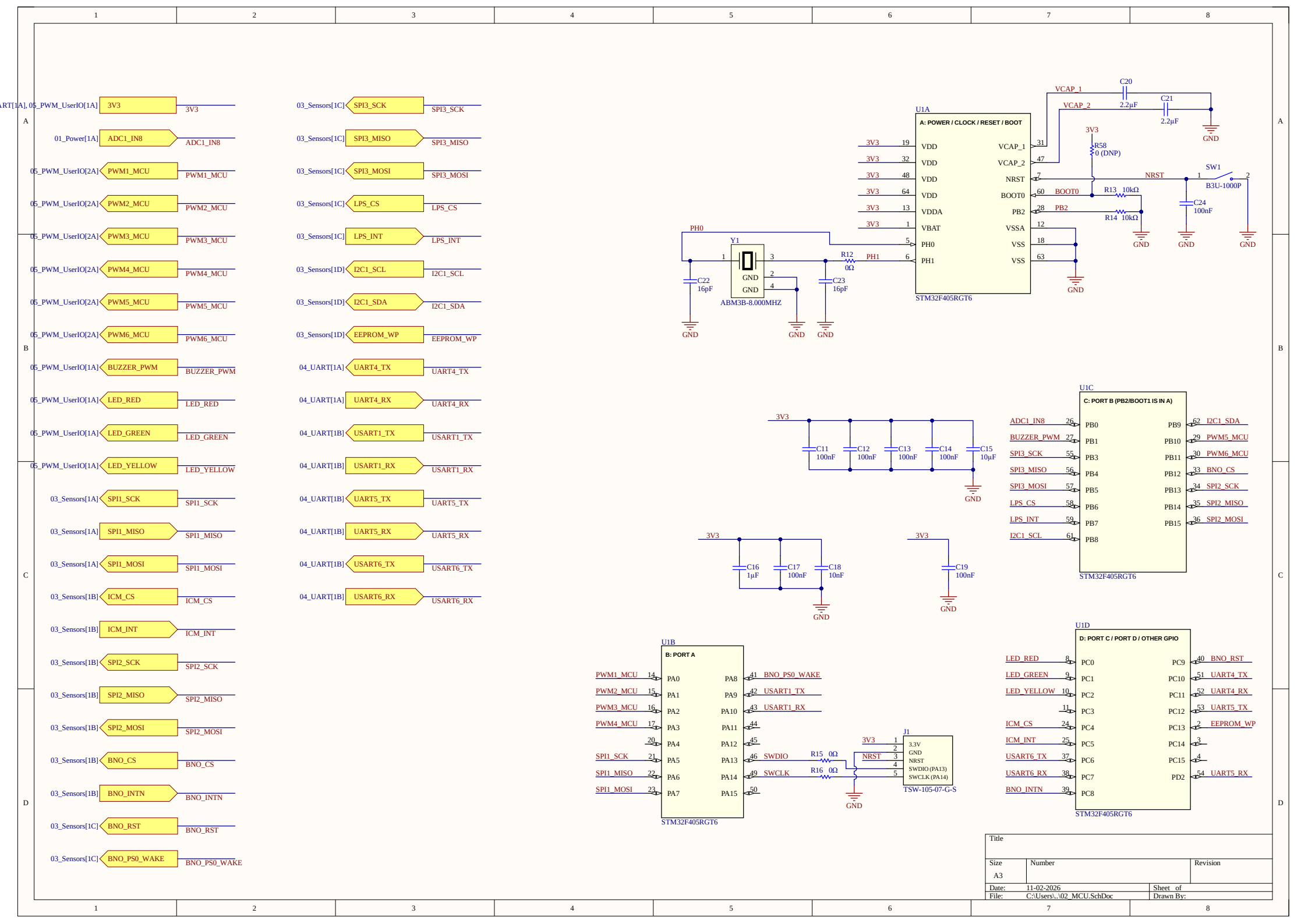
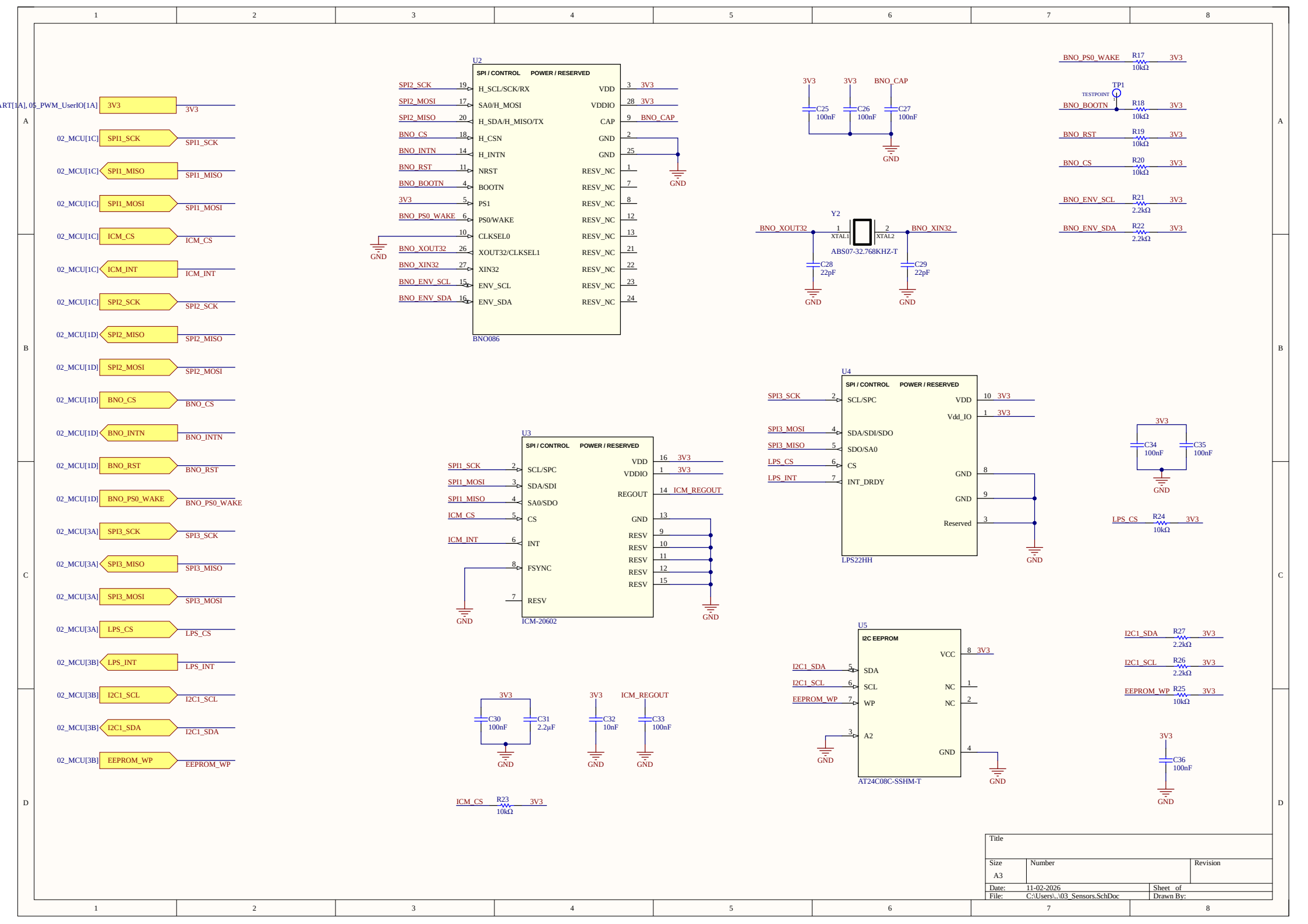
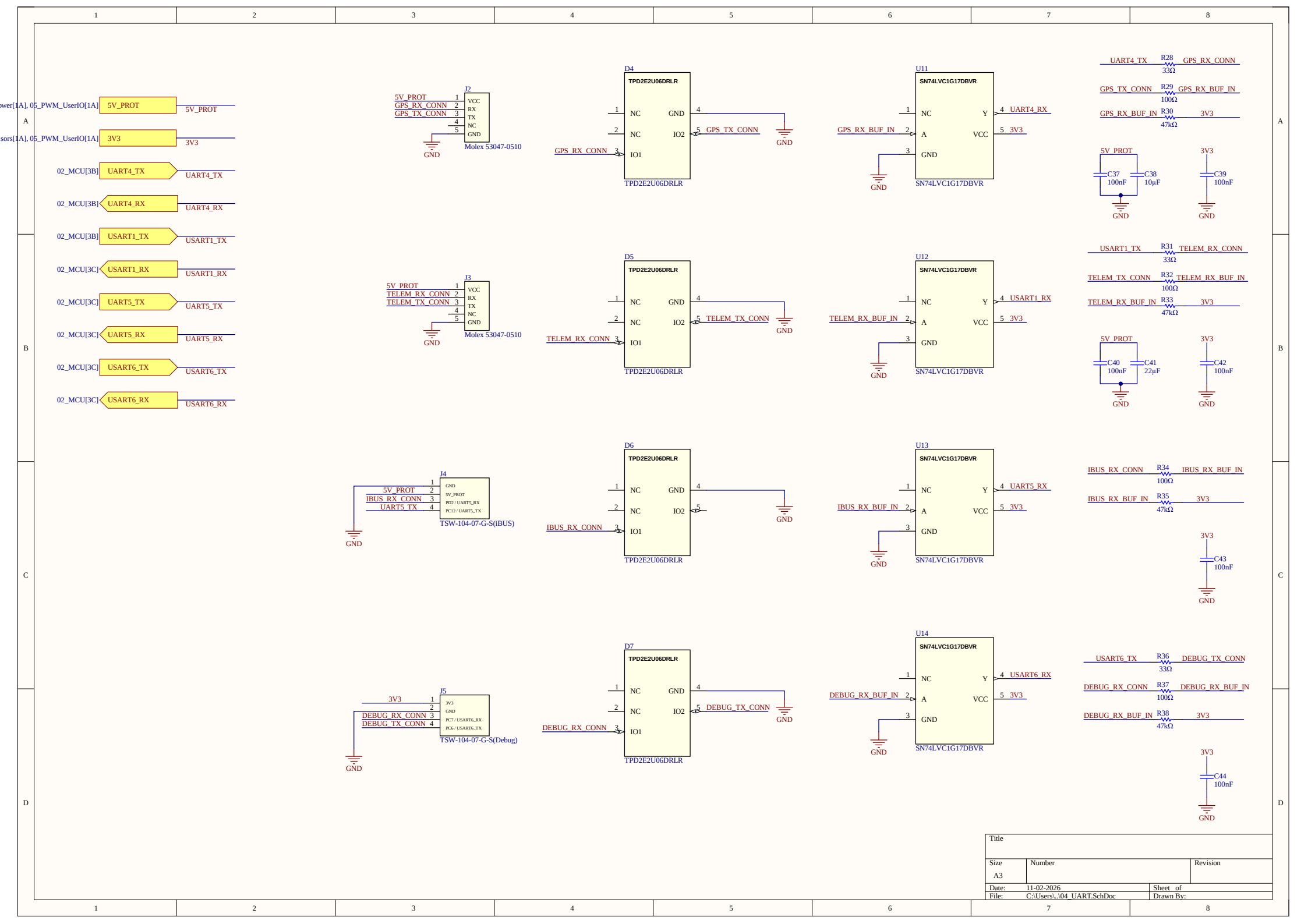


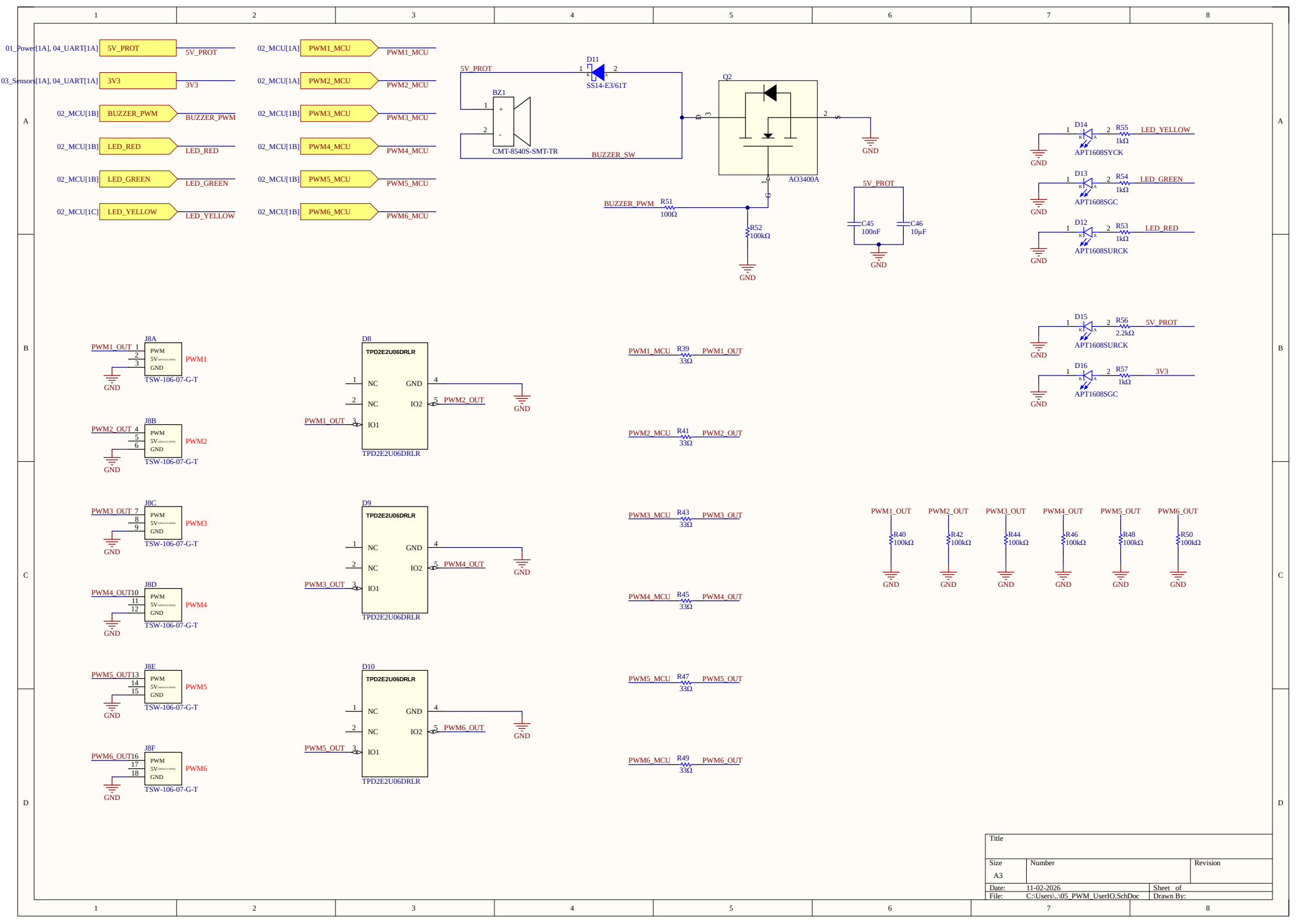
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Size A3	Number	Revision
Date: 11-02-2026		Sheet of
File: C:\Users\..\01_Power.SchDoc		Drawn By:



Title		
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File: C:\Users\..\02_MCU.SchDoc		Drawn By:


ART[1A], 05_PWM_UserIO[1A]
3V3
02_MCU[1C] SPI1_SCK
02_MCU[1C] SPI1_MISO
02_MCU[1C] SPI1_MOSI
02_MCU[1C] ICM_CS
02_MCU[1C] ICM_INT
02_MCU[1C] SPI2_SCK
02_MCU[1D] SPI2_MISO
02_MCU[1D] SPI2_MOSI
02_MCU[1D] BNO_CS
02_MCU[1D] BNO_INTN
02_MCU[1D] BNO_RST
02_MCU[1D] BNO_PS0_WAKE
02_MCU[3A] SPI3_SCK
02_MCU[3A] SPI3_MISO
02_MCU[3A] SPI3_MOSI
02_MCU[3A] LPS_CS
02_MCU[3B] LPS_INT
02_MCU[3B] I2C1_SCL
02_MCU[3B] I2C1_SDA
02_MCU[3B] EEPROM_WP
U2
SPI / CONTROL
POWER / RESERVED
SPI2_SCK 19 H_SCL/SCK/RX
SPI2_MOSI 17 SA0/H_MOSI
SPI2_MISO 20 H_SDA/H_MISO/TX
BNO_CS 18 H_CSN
BNO_INTN 14 H_INTN
BNO_RST 11 NRST
BNO_BOOTN 4 BOOTN
3V3 5 PS1
BNO_PS0_WAKE 6 PS0/WAKE
10 CLKSEL0
BNO_XOUT32 26 XOUT32/CLKSEL1
BNO_XIN32 27 XIN32
BNO_ENV_SCL 15 ENV_SCL
BNO_ENV_SDA 16 ENV_SDA
VDD 3 3V3
VDDIO 28 3V3
CAP 9 BNO_CAP
GND 2
GND 25
RESV_NC 1
RESV_NC 7
RESV_NC 8
RESV_NC 12
RESV_NC 13
RESV_NC 21
RESV_NC 22
RESV_NC 23
RESV_NC 24
GND
BNO086
3V3
3V3
BNO_CAP
C25 100nF
C26 100nF
C27 100nF
GND
BNO_PS0_WAKE
R17 10kΩ 3V3
TP1
TESTPOINT
BNO_BOOTN
R18 10kΩ 3V3
BNO_RST
R19 10kΩ 3V3
BNO_CS
R20 10kΩ 3V3
BNO_ENV_SCL
R21 2.2kΩ 3V3
BNO_ENV_SDA
R22 2.2kΩ 3V3
Y2
BNO_XOUT32
BNO_XIN32
XTAL1
XTAL2
ABS07-32.768KHZ-T
C28 22pF
C29 22pF
GND
GND
U3
SPI / CONTROL
POWER / RESERVED
SPI1_SCK 2 SCL/SPC
SPI1_MOSI 3 SDA/SDI
SPI1_MISO 4 SA0/SDO
ICM_CS 5 CS
ICM_INT 6 INT
8 FSYNC
7 RESV
VDD 16 3V3
VDDIO 1 3V3
REGOUT 14 ICM_REGOUT
GND 13
RESV 9
RESV 10
RESV 11
RESV 12
RESV 15
GND
GND
ICM-20602
U4
SPI / CONTROL
POWER / RESERVED
SPI3_SCK 2 SCL/SPC
SPI3_MOSI 4 SDA/SDI/SDO
SPI3_MISO 5 SDO/SA0
LPS_CS 6 CS
LPS_INT 7 INT_DRDY
VDD 10 3V3
Vdd_IO 1 3V3
GND 8
GND 9
Reserved 3
GND
LPS22HH
3V3
C34 100nF
C35 100nF
GND
LPS_CS
R24 10kΩ 3V3
3V3
C30 100nF
C31 2.2µF
GND
3V3
C32 10nF
GND
ICM_REGOUT
C33 100nF
GND
ICM_CS
R23 10kΩ 3V3
U5
I2C EEPROM
I2C1_SDA 5 SDA
I2C1_SCL 6 SCL
EEPROM_WP 7 WP
3 A2
VCC 8 3V3
NC 1
NC 2
GND 4
GND
GND
AT24C08C-SSHM-T
I2C1_SDA
R27 2.2kΩ 3V3
I2C1_SCL
R26 2.2kΩ 3V3
EEPROM_WP
R25 10kΩ 3V3
3V3
C36 100nF
GND
Title
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ower[1A], 05_PWM_UserIO[1A]
5V_PROT
5V_PROT
sors[1A], 05_PWM_UserIO[1A]
3V3
3V3
02_MCU[3B]
UART4_TX
UART4_TX
02_MCU[3B]
UART4_RX
UART4_RX
02_MCU[3B]
USART1_TX
USART1_TX
02_MCU[3C]
USART1_RX
USART1_RX
02_MCU[3C]
UART5_TX
UART5_TX
02_MCU[3C]
UART5_RX
UART5_RX
02_MCU[3C]
USART6_TX
USART6_TX
02_MCU[3C]
USART6_RX
USART6_RX
J2
5V_PROT
GPS_RX_CONN
GPS_TX_CONN
VCC
RX
TX
NC
GND
Molex 53047-0510
GND
D4
TPD2E2U06DRLR
NC
NC
IO1
GND
IO2
GPS_TX_CONN
GPS_RX_CONN
GND
TPD2E2U06DRLR
U11
SN74LVC1G17DBVR
NC
A
GND
Y
VCC
UART4_RX
3V3
GPS_RX_BUF_IN
GND
SN74LVC1G17DBVR
UART4_TX
R29
GPS_RX_CONN
33Ω
GPS_TX_CONN
R29
GPS_RX_BUF_IN
100Ω
GPS_RX_BUF_IN
R30
3V3
47kΩ
5V_PROT
C37
100nF
C38
10µF
GND
3V3
C39
100nF
GND
J3
5V_PROT
TELEM_RX_CONN
TELEM_TX_CONN
VCC
RX
TX
NC
GND
Molex 53047-0510
GND
D5
TPD2E2U06DRLR
NC
NC
IO1
GND
IO2
TELEM_TX_CONN
TELEM_RX_CONN
GND
TPD2E2U06DRLR
U12
SN74LVC1G17DBVR
NC
A
GND
Y
VCC
USART1_RX
3V3
TELEM_RX_BUF_IN
GND
SN74LVC1G17DBVR
USART1_TX
R31
TELEM_RX_CONN
33Ω
TELEM_TX_CONN
R32
TELEM_RX_BUF_IN
100Ω
TELEM_RX_BUF_IN
R33
3V3
47kΩ
5V_PROT
C40
100nF
C41
22µF
GND
3V3
C42
100nF
GND
J4
5V_PROT
IBUS_RX_CONN
UART5_TX
GND
5V_PROT
PD2 / UART5_RX
PC12 / UART5_TX
TSW-104-07-G-S(iBUS)
GND
D6
TPD2E2U06DRLR
NC
NC
IO1
GND
IO2
IBUS_RX_CONN
GND
TPD2E2U06DRLR
U13
SN74LVC1G17DBVR
NC
A
GND
Y
VCC
UART5_RX
3V3
IBUS_RX_BUF_IN
GND
SN74LVC1G17DBVR
IBUS_RX_CONN
R34
IBUS_RX_BUF_IN
100Ω
IBUS_RX_BUF_IN
R35
3V3
47kΩ
3V3
C43
100nF
GND
J5
3V3
DEBUG_RX_CONN
DEBUG_TX_CONN
3V3
GND
PC7 / USART6_RX
PC6 / USART6_TX
TSW-104-07-G-S(Debug)
GND
D7
TPD2E2U06DRLR
NC
NC
IO1
GND
IO2
DEBUG_TX_CONN
DEBUG_RX_CONN
GND
TPD2E2U06DRLR
U14
SN74LVC1G17DBVR
NC
A
GND
Y
VCC
USART6_RX
3V3
DEBUG_RX_BUF_IN
GND
SN74LVC1G17DBVR
USART6_TX
R36
DEBUG_TX_CONN
33Ω
DEBUG_RX_CONN
R37
DEBUG_RX_BUF_IN
100Ω
DEBUG_RX_BUF_IN
R38
3V3
47kΩ
3V3
C44
100nF
GND
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